

Nicole Leow Ke Xin

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Summary

Highly motivated and detail-oriented Mechatronics Engineering and Computer Science student at UNSW with hands-on experience in full-stack software development, data analysis and cross-team collaboration. Through industry and student project roles, I have designed and built software features using modern frontend and backend technologies, worked with databases and internal systems, and used reporting and analysis tools to support operational and engineering teams. I bring strong analytical thinking, clear communication skills and a proactive approach to learning, and I enjoy working in fast-paced environments. I am keen to continue developing my technical and professional skills while contributing meaningfully to a team.

Education

UNSW Sydney (2021 - present)

- Double degree in Bachelor of Mechatronics Engineering (Honours) and Bachelor of Science (Computer Science)
- High Distinction WAM
- Awarded the International Scientia Coursework Scholarship for 100% tuition fee remission

Sunway College (2020 - 2021)

- 4A* (High Distinction) in the Cambridge A Levels Physics, Chemistry, Biology and Maths

Technical Skills

- Fusion 360
- SolidWorks
- 3D printing and Laser Cutting
- C
- C#
- C ++
- Java
- Python
- Javascript
- SQL
- React
- HTML & CSS
- Microsoft Office

Experiences

UNSW Casual Academic (2024 - Present)

- Tutor for five undergraduate courses in programming (OOP, Java, C++) and engineering.
- Managed and led classes of engineering students, guiding them through core concepts and practical applications.
- Fostered a supportive and interactive learning environment.

Fullstack Software Engineer Intern (2025 - Present)

- Developed and maintained full-stack features using C#, TypeScript and React.
- Built and maintained backend services using .NET, Entity Framework and LINQ, integrating with relational databases.
- Wrote and optimised SQL queries to support application features and data workflows.
- Collaborated closely with designers, engineers and the product manager to plan features, refine requirements and deliver project milestones.

- Participated in sprint planning, code reviews and team discussions to improve code quality and delivery processes.

Telstra Undergraduate Domain Engineer (2024 - 2025)

- Gained expertise in Telstra's Distributed Production Network, focusing on its structure and functionality.
- Analysed and corrected data errors within the Digital Video Network, identifying issues and implementing solutions.
- Collaborated with teams to update data processing and error generation SQL scripts, ensuring accurate data flow between systems.

Jane Street Insight Program Participant (2024)

- Engaged in intensive learning sessions covering probability, market structure, and arbitrage, including interactive group activities and mock trading simulations.
- Gained valuable insights into the dynamic world of finance.

UNSW Mechatronics Society Executive Member (2023)

- Ensured the smooth operation of the society by overseeing events, workshops, and projects run by the society.
- Worked collaboratively with other executive members to develop and execute strategies to promote the society and attract new members.
- Coordinated with external stakeholders and professionals to establish partnerships and secure funding.

Connected Health Software Engineer (2023)

- Responsible in developing machine learning modules for statistical analysis to improve the accuracy of image-guided surgical tool in the removal of cancerous oral lesions.
- Implement software using programming languages such as Julia and MATLAB.

UNSW Mechatronics Society Projects Subcommittee (2022)

- Responsible in planning, organising, and running workshops and projects.
- 2022 UNSW SumoBots Competition – Run and assisted in various workshops to guide competitors through designing and building a sumo robot. Led workshops in CAD, 3d Printing and coding using software like SolidWorks and Cura and Arduino IDE.